

Challenges of machine learning-based RUL prognosis: A review on NASA's C-MAPSS data set

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Abstract—The estimation of a system's or a component's remaining useful life (RUL) is considered the most complex task in predictive maintenance, at the same time the most beneficial one. In this brief review paper, we survey the state-of-the-art in machine learning-based RUL prognosis based on research on NASA's C-MAPSS data set. We identify the frequently used models, comparatively evaluate model performance and survey the used feature extraction methods. As a main contribution, we formulate challenges in the field, independently of the C-MAPSS data set. Among the challenges are interpretability, model uncertainty and domain adaptation, i.e. transfer learning. The identified challenges may serve to identify potential research directions, in order to further push the field of machine learning applied to RUL prognosis.

Index Terms—machine learning, remaining useful life, predictive maintenance, time series, interpretability

I. INTRODUCTION

Modern machinery is becoming increasingly complex to enable faster and flexible operation. In order to make optimal use of machines and to ensure planning security, the failure-free execution of an intended task is more important than ever. With the ever growing digitization of fields like manufacturing [1], automotive [2]–[4] and aviation [5], predictive maintenance (PdM) has gained a lot of attention. At the same time, machine learning (ML) has made its way from research to industrial applications. Advanced ML methods, e.g. deep learning (DL), have achieved unprecedented accuracies [6]. Data-driven PdM, specifically ML, is now widely used for the detection, diagnosis and prognosis of faults. The prognostics task to predict an asset's remaining useful life (RUL), i.e. the time span from the current time point to the end of the useful life [7], is considered as the most complex part of PdM [8], yet the most beneficial one. Predicting the time of failure enables not only cost-effective scheduling of maintenance, but also related activities like spare part provisioning [9], [10].

In this paper, we identify the challenges for ML-based RUL prognosis from recently published work and discuss possible research directions. Due to the high relevance of the field, the number of publications is quite high – an initial search on Scopus revealed > 6000 publications. To gain a thorough, fair and representative overview, we opted for a systematic literature review from the perspective of the widely used NASA Commercial Modular Aero-Propulsion System

Simulation (C-MAPSS) turbofan engine data set [11]. From the review we deduce general challenges for ML-based RUL prognosis, transferable to other domains.

There is a number of reasons for the choice of the C-MAPSS data set: C-MAPSS is recognized as the benchmark data set for RUL prognosis [12]–[15]. Furthermore, its properties allow to relate the findings to a variety of other use cases in research and industry: (a) the data set contains multivariate time series complemented by metadata with operational settings, (b) there are multiple physical reactions between the recorded properties [11], [16], (c) there are multiple fault modes and operating conditions, and (d) manufacturing variations, different lifespans, and natural failures are considered.

This paper makes the following contributions: (1) We systematically review research on NASA's C-MAPSS data set, thereby surveying the state-of-the-art in RUL prognosis. (2) From the review we rank ML models based on how frequently they were used as well as their model performance – showing that the best performing models are not the ones that were most frequently used. (3) We identify challenges, independent of C-MAPSS. These may serve as potential research directions.

II. RELATED WORK AND SCIENTIFIC CONTRIBUTION

Multiple reviews of ML-based PdM have been conducted in the recent years: for general PdM [17]–[19], for a specific application such as bearings [20], [21] or ships [22], or for a specific method such as convolutional neural networks [23] or Bayesian networks [24]. Both, systematic literature reviews [25], [26] and other type of reviews [18], [27] can be found. Furthermore, reviews specific to the C-MAPSS data set have been conducted in the past, most notably [12], [28] and most recently [29]. As the C-MAPSS data set is widely recognized as a benchmark data set, those reviews aim to further the benchmark cause, by providing an overview and taxonomy specific to the C-MAPSS data set. We argue that by using the C-MAPSS data set as an example domain, we derive general challenges that apply to PdM. Thus, our main focus is not on the data set, but on its properties and their implications for data-driven approaches.

Therefore, this work aims to bridge the gap between application-specific reviews and reviews of general PdM by using the widely accepted benchmark data set C-MAPSS [12]

TABLE I
C-MAPSS DATA SETS OVERVIEW (FD001 ... FD004), WHERE RMSE
REFERS TO THE ROOT-MEAN-SQUARE ERROR

	FD001	FD002	FD003	FD004
Train/Test data	100/100	260/259	100/100	249/248
Operating Conditions	1	6	1	6
Fault Modes	1	1	2	2
Minimum RMSE	8.78 [21]	12.25 [21]	7.11 [35]	13.19 [21]
Median RMSE	14.36	21.59	14.95	23.89
Maximum RMSE	30.95	104.58	30.55	35.6

[23] to systematically demonstrate what challenges arise for ML-based PdM, specifically RUL prognosis.

III. C-MAPSS DATA SET

The data set generated by the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) software of the NASA is selected as the exemplary data set, for which current RUL approaches and challenges will be examined. The data set consists of multiple turbofan engines, which are typically used for passenger aircraft vehicles. Such engines are powered by gas and use the energy resulting from the combustion of the gas to power a fan, which propels the air backwards and thereby achieve forward thrust [30].

The data set results from simulations and consists of 21 different sensor measurements (i.e. multivariate time series) across four subsets (FD001...FD004) and an additional competition subset [11]. Each of the data sets has a 24-dimensional vector with three operational settings as well as the 21 sensor signals. The goal is to predict the RUL of the engines. Each of the subsets has its own predefined train and test set [28] (see Table I). Each engine starts at a random healthy state and begins deteriorating after a short time period, therefore the engines have different lifespans [31]. The engines are considered to be influenced by manufacturing variations and therefore exhibit slightly different characteristics between one another [11]. In the training subset, sensor measurements are available until end of life of the engine, whereas in the test subset, measurements stop some time before the failure [32]. The label in the training set is the time point of the failure [33]. Even though the data set is simulated, this is done in accordance with real use-cases where the exact RUL, apart from the point of failure, can only be estimated [34]. For the test subsets, a single label specifies the RUL after the last measurement [11]. As the data set is the most popular of the NASA data sets and considered as the benchmark data set for RUL prognosis [12], it is often the first choice for benchmarking new prognostic approaches.

IV. REVIEW METHODOLOGY

To gain a thorough, fair and representative overview of research on the C-MAPSS data set, a systematic literature review is conducted, as per the methodology of [36]. Realizations for this type of review can also be found in e.g. [25], [26]. The protocol used for the review is given in Table II. The search was conducted Dec 2020 on ScienceDirect (SD), IEEEExplore,

TABLE II
REVIEW METHODOLOGY

Search strategy:	
SS1	Initial search: search terms in scientific databases
SS2	Search for additional literature based on secondary literature
SS3	Backward search in the identified literature
Search terms (SS1):	
	(Data driven OR Machine Learning OR ML OR Deep Learning OR DL) AND (PHM OR Predictive Maintenance OR Remaining Useful Life OR RUL OR Prognosis) AND (Turbofan OR *MAPSS)
Exclusion criteria:	
E1	publication date before 2010
E2	no experiment conducted
E3	secondary literature, i.e. reviews, etc.
E4	use of proprietary software
E5	use of metrics other than RSME or use of custom test sets

TABLE III
SYSTEMATIC SEARCH RESULTS. AFTER AN INITIAL SEARCH, PAPERS WERE EXCLUDED (-) USING THE EXCLUSION CRITERIA AND ADDED (+) BASED ON SECONDARY LITERATURE AND A BACKWARD SEARCH.

Database	SD	IEEE	ACM	MDPI	SL	Total
initial search (SS1,E1)	136	61	19	5	35	256
- duplicates	0	2	2	0	2	6
- due to title (E2-E4)	38	8	9	0	24	79
- due to abstract (E2-E4)	26	7	4	1	2	40
- due to content (E2-E4)	22	11	4	0	2	39
+ additional search (SS2,SS3)	6	3	0	3	4	16
- not comparable (E5)	13	9	1	1	3	27
Selected publications	43	25	1	6	6	81

ACM, MDPI and SpringerLink (SL) and 81 publications were selected. The search results are shown in Table III.

V. ANALYSIS OF REVIEW RESULTS

A. Used ML methods

The majority of the 81 reviewed papers employ an artificial neural network (ANN) based approach to solve the prognostic task. More precisely 64 publications employ some sort of ANN, whereas 17 approaches make use of other ML or probabilistic methods. An overview is given in Table IV. Recurrent Neural Networks (RNN) and its variants, such as Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU) or Echo State Networks (ESN), make up the majority of the ANN based approaches. Convolutional Neural Networks (CNN) are also very popular, whether used on their own or in combination with LSTMs. Feedforward neural networks in their basic form of multi-layer perceptron (MLP) and more advanced architectures like extreme learning machines (ELM) are not as widely used. Tree-based methods, including ensembles of trees, are the most popular of the other ML methods, followed by similarity-based methods. Furthermore, the category *other* and *other ANN* contain all approaches which were used in one of the surveyed papers each.

Degradation is usually thought of as a slow process over time, for which LSTMs, GRUs and ESNs excel because of their ability to model long-term dependencies. In contrast to that, fault diagnosis can be considered as a pattern recognition problem, for which long-term dependencies are not as impor-

TABLE IV
OVERVIEW OF THE USED MODELS

Model	Publications	Count
LSTM	[14], [35], [37]–[62]	28
CNN+LSTM	[21], [33], [63]–[70]	10
CNN	[13], [31], [71]–[78]	10
ESN, GRU, RNN	[79]–[85]	7
Tree-based	[32], [86]–[90]	6
Other ANN	[15], [91]–[95]	5
Other	[96]–[100]	5
Similarity-based	[101]–[104]	4
MLP, ELM	[105]–[108]	4
Particle Filter	[109], [110]	2

tant as specific patterns. In [111] DL approaches for general PdM are surveyed, stating that CNNs are primarily used for fault diagnosis and detection due to their ability of local feature extraction. The use of DL methods for RUL are surveyed in [19]. They mention that CNNs achieved great results in fault diagnosis, but are not as often used in prognostics, but they do not give a reason as to why. Furthermore, they found much more applications of RNNs, especially LSTMs which is in accordance with our review. The authors of [112] survey the application of data-driven models in Prognostics and Health Management (PHM) and came to the conclusion that RNNs are the most widely used ANN architecture for prognosis tasks. Similar results can be deduced from this systematic review, as RNNs make up 43% of the surveyed approaches. However, this does not infer that RNNs do necessarily perform best, as discussed in Section VI-B.

B. Feature Extraction and Selection

Most publications make use of feature selection as 61 of the publications select features either based on correlation, but most often on prognostic indicators. The most used prognostic indicator was monotonicity followed by trendability, used 30 and 27 times, respectively. Other methods for feature selection include LASSO [86] and variance threshold selection [74], however, they were not widely used.

In terms of feature processing, out of the identified publications, 43 do not use any feature processing methods apart from normalization. The remaining publications that use additional feature processing methods can be loosely categorized into five different categories, namely (a) dimensionality reduction, (b) clustering, (c) noise filtering, (d) statistical and signal processing based feature extraction and (e) feature learning. Nine of the remaining 38 publications use clustering to discover additional features where k-means was used in eight of them. Dimensionality reduction is proposed eight times, exclusively in the form of (Kernel) Principal Component Analysis. Furthermore, various filters were applied for the purpose of noise reduction, such as Gaussian [102], Kalman [100], or exponential filters [80]. 18 approaches extracted features with statistical or signal processing methods. Among those, no particular trend can be noticed. Lastly 14 approaches make use of an explicit model for feature learning. Among those methods, autoencoders (AE) were the most popular, e.g. sparse AE [49], GRU AE [83] or variational AE [97].

VI. CHALLENGES

In this section we discuss observations from the surveyed papers and deduce a number of challenges. While these challenges are deduced for research on the C-MAPSS, they can be generalized to ML for predictive maintenance.

A. Quantitative comparability

Unfortunately, even though test sets are readily provided, some authors decided against the use of those test sets and instead use custom test sets or reduce the provided test sets to specific engines. Furthermore, many different performance measures are used. For example, some publications use the mean absolute error or the mean average percentage error or even a customary metric. However, most publications adopted a de-facto standard of using the Root-Mean-Square Error (RMSE) as a performance metric. Another widely spread performance metric is the competition score which was proposed for the PHM 2008 competition. It is a custom score which, in contrast to RMSE, penalizes late predictions more than early predictions. Using a symmetric loss function may not reflect the impact of a too early or too late prediction for a given application. Thus, even though this data set is considered as a benchmark data set, it is difficult to compare some approaches, using different metrics or modified test sets, against each other.

Challenge 1: Limited quantitative comparability. Quantitative comparability is limited for published research, preventing an overall view over all published approaches. A superior approach might therefore be overlooked. For research on the C-MAPSS data set, the following reasons were identified: (1) the use of different error metrics, (2) the use of test sets other than the pre-defined ones, (3) some approaches were not applied to all of the pre-defined data sets. We underline the importance of research applying or proposing methods for new problem settings, where the data is possibly not publicly available. In addition to that, method comparison using benchmark data is desirable, hence, suggesting to report results in a comparable way for that case.

B. Predictive Performance

The distribution of the comparable approaches with the RMSE metric are shown in Table I. Comparing the papers' performances on the different data sets of C-MAPSS, it becomes obvious that multiple operating conditions are much more challenging than single operating conditions (FD002 and FD004 have six operating modes, as opposed to FD001 and FD003 with one). Furthermore, multiple operating conditions seems much more challenging, than the presence of multiple fault modes. It is observable, that the lowest median error is achieved on the FD001 data set, whereas the lowest overall error is achieved on the FD003 data set. Contrary to that the highest median error is found on FD004 and the highest absolute error on FD002. A further interesting observation is that the 25%-quantil is lower on FD003 compared to FD001 even though FD003 is a more complicated data set due to the addition of a second fault mode. However, the interquartile

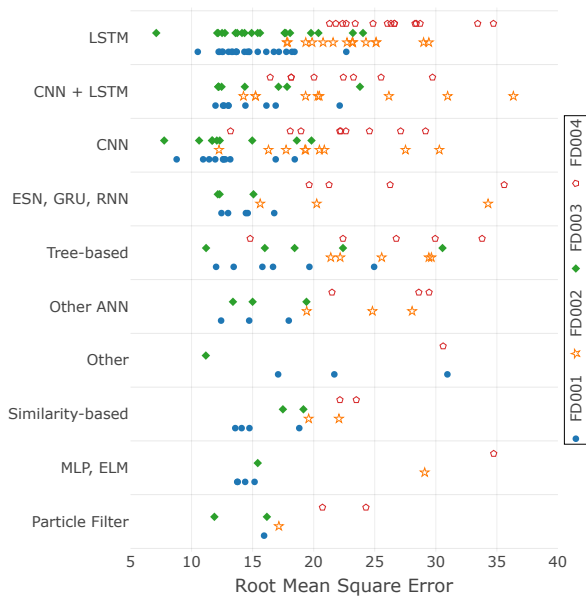


Fig. 1. RMSE scores across models and data sets

range is also larger on FD003 hinting at a bigger difference between the approaches on FD003.

Challenge 2: Higher error rates for multiple operating conditions. An increase in the number of operating conditions leads to a significant increase in the error rates. Since in machinery regularly multiple operating conditions are encountered, this calls for research in this direction. Additional benchmark data sets with a varying number of operating conditions are desirable.

On the FD003 set more approaches achieve very low error rates compared to the FD001 set (see Fig. 1). At the same time more models have higher error rates on FD003 compared to FD001. The better median RMSE on FD003 set can be attributed to papers using CNNs and LSTMs which overall tend to have lower error rates. Furthermore, the traditional ML methods show a higher variance (see e.g. tree-based vs. LSTM) and perform worse on average compared to ANN approaches. Still, traditional methods do not necessarily perform badly. The three best performing traditional approaches used Particle Filter (PF) [110], random forests [86] and Fractional Brownian Motion (FBM) [99].

Three ANN approaches outperformed the aforementioned traditional approaches: two CNNs and one LSTM. The 1-D CNN proposed in [68] uses stacked separable convolutions and residual connections. An adaptive spatio-temporal graph CNN, constructed from a dynamic graph neural network and a 1-D CNN with dilated convolutions, was used in [77]. A dual LSTM framework was proposed in [35]. With that approach the change point, i.e. the point in time at which degradation starts, is first predicted by a first LSTM, and based on that the RUL prediction conducted by a second LSTM. The approach achieves the best prediction performance on stable operating conditions but multiple fault modes (FD003).

For FD002 and FD004, i.e. multiple operating conditions, a worse performance on average across all model types can be observed. In Fig. 1, we can see no model type achieving better scores on FD002/FD004 as compared to FD001 and FD003. Traditional ML approaches perform worse on average than ANN methods, when multiple operating conditions are present. For FD002, the only approach competitive with ANNs is the similarity-based PF approach in [109], where similar engines are selected via maximum mean discrepancy and the kernel two sample test. Furthermore, for FD002 and FD004, traditional ML methods have been used less frequently, indicating that ANN based methods seem to be more popular when facing multiple operating conditions – their performances are generally better on average. A particularly interesting observation is the lack of LSTMs among the best performing models for multiple operating conditions. Even though they are very widely used, they seem to be outperformed by CNNs as well as CNN and LSTM combination approaches. The top three approaches for FD002 are a CNN and combinations of CNNs and LSTMs. In [72] a network with two parallel paths, a 1D-CNN and an LSTM path, was proposed. The paths are fused with three fully-connected layers. The approach was further improved by [73] through the addition of a GRU path, thereby creating a network with a GRU, a CNN and an LSTM path. The best performing model is the graph CNN proposed by [77].

To summarize, ANN based approaches do generally perform better across all subsets of the C-MAPSS data set. However, there are some traditional approaches which remain competitive, albeit if predictive performance is the primary goal there is hardly a way around CNNs and LSTMs. The C-MAPSS data set is not saturated in terms of the predictive performances of the surveyed approaches. Hence, there is still a margin left across all subsets, to achieve perfect prediction results. This is also shown by the fact that only 18 of the 81 publications state other reasons than performance as their motivation. We argue that the implementability of an approach in a real-world use case does not solely depend on model performance.

Challenge 3: Research focus on model performance. The majority of the research is motivated by the minimization of error rates. While high error rates make a method unusable, we argue that regarding implementability, additional properties like interpretability, scalability, robustness, uncertainty measures and others should also be considered.

C. Interpretability

Only three of the 81 approaches state that they were explicitly motivated by interpretability. Four additional publications were identified that use an intrinsically interpretable method. An unfortunate observation is that all but two of the publications just use a single subset. Interpretability as a primary motivation is restricted to LSTM-based approaches, thus the three publications which explicitly state interpretability as a goal all employ LSTMs. The first approach [39] uses a bidirectional LSTM with attention mechanism. The weights of the attention mechanism are visualized to highlight the

timesteps with the highest importance. The second approach [96] incorporates an LSTM as a black-box predictor but use a Wiener process as a surrogate model to quantify uncertainty. Specifically, the Wiener process is trained to approximate the predictions of the LSTM and provide the ability to quantify uncertainty due to its stochastic nature. The authors of [47] proposed the last ANN-based interpretable approach. They use a structured-effect model which splits RUL prediction into three components, namely an LSTM, a non-parametric baseline and a covariate-based prediction. The non-parametric baseline is based on the weibull distribution and the covariate-based prediction is obtained via ordinary least squares, i.e. it is a linear component. The LSTM is trained to account for the remaining variations of the degradation, i.e. for the variance that can not be explained by the other two components.

Out of the four traditional methods, two use similarity measures. The one proposed in [102] performed much better in terms of predictive performance. Instead of using the global similarity of the degradation trajectories, they divide the series into multiple segments and extract local degradation patterns via dynamic time warping. In [101] the similarity of the trajectories are used. They use kernel density estimation to model the probability density function with the goal to obtain uncertainty estimates. Hence, their prediction is associated with a confidence interval. The remaining two of the interpretable approaches use PF [109] and FBM [99], which are both considered as interpretable. Parallels can be drawn to physical modeling in which both PF and FBM are extensively used. The implementations are, however, purely data-driven in these cases. To summarize, it can be observed that the majority of the ANN approaches lack interpretability. Thus, it is hindering that a lot of the interpretable publications did not use all subsets. However, apart from [99], they did not reach the predictive performances of other ML and DL-based approaches. Generally speaking, interpretability seems to be an open challenge for multivariate time series.

Challenge 4: Interpretability of ML models. Interestingly, while the majority of research uses ANNs – which are commonly referred to as black box models – the interpretability of these models is only addressed in very few publications (see e.g. [113] for a general overview). We argue that since maintenance leads to costly repairs or replacement of parts, users will demand to understand RUL decisions. Hence, the research field of interpretable ML is likely to become more important for RUL and possibly be a decisive factor.

D. Quantification of Uncertainty

The most emphasized motivation besides performance, is uncertainty as eight publications mention it as their primary motivation. Additionally, many publications implicitly incorporate both uncertainty estimates and interpretability. For example [96] and [99] did not actively specify interpretability as a goal, but rather uncertainty estimation. However, due to the nature of their approach, both are combined.

Similar to interpretability, uncertainty estimation is often mentioned together with ANN based approaches as (deep)

ANNs generally lack both, interpretability and the ability to provide uncertainty estimations for their point predictions. Six publications were motivated to provide the ability of uncertainty estimation for ANNs. Thereby the proposed methods range from the use of special activation functions by [56] to the use of Gaussian Process Regression by [93], to the surrogate Wiener process in [96] and finally to the use of Bayesian methods in [15] and [37]. Furthermore publications exist which provide measures of uncertainty but are not explicitly motivated by it such as [110]. The use of Bayesian methodology is especially prevalent for the use in neural networks, which ultimately manifests itself in the creation of Deep Bayesian Neural Networks such as in [15]. Non ANN-based approaches explicitly select probabilistic or stochastic models that naturally provide an uncertainty quantification such as FBM in [99] or PF in [109].

Challenge 5: Reporting model uncertainty. Reporting model uncertainty is viewed as important in a number of publications. We believe it is essential for a real-world use case, where the data might show different properties from what the model was trained on, termed concept drift [114]. This is important to accurately assess if a model still operates as intended.

E. Domain Adaptation and Transfer Learning

Domain adaptation concerns itself with the transfer of knowledge. On the C-MAPSS data set, domain adaptation is performed via the different subsets, e.g. a model is trained on FD001 and evaluated on FD002. We noticed publications tackling this problems in two ways: Either by training on a easier subset (one operating condition) and transferring to a more difficult subset (multiple operating conditions) or vice versa. The choice of models for was limited to neural networks and consists of a CNN [64], LSTMs [38], [51], [58] as well as a CNN-LSTM combination approach [76]. [87] used self organizing models for feature extraction and combined them with random forest regression. They achieve domain adaptation via the self organizing models. The best domain adaptation performance was achieved by [38] who used an LSTM in an unsupervised way with adversarial learning.

Challenge 6: Domain adaptation. The problem of real-world data following a different distribution compared to the training data is recognized by a number of publications. The different subsets of the C-MAPSS data allow to test domain adaptation. This might be an inspiration for other data sets to also offer different subsets.

F. Data Size

A key difference between the simulated C-MAPSS data set and real-world data is the size. C-MAPSS has hundreds of run-to-failure degradation trajectories. As degradation is a slow process, gathering such amounts of data is very time-consuming or requires ways to accelerate the aging process and possibly a large assortment of the same system. Furthermore, letting operational equipment run to failure is not an option for all use-cases. In addition, acquiring labels is often paired with significant costs, e.g. due to required expert

knowledge, or might even be impossible as the condition can not be observed. Some publications therefore propose an unsupervised [40] or semi-supervised [97] setting. Most approaches, specifically the predominantly used ANNs, benefit from the large quantity of data available. The performance of the models would most likely change with less data, in favor of more shallow models. Only one of the publications was motivated by using a reduced training data set [105].

Challenge 7: Need for large amounts of labelled data. The research on C-MAPSS relies on the large amount of available data, which is in addition labelled. For many real-world applications such a setting is not given. Research directions can be methods requiring less data (up to one-shot learning [115]) and efficient means of data labelling [116]–[118].

CONCLUSION

We systematically reviewed research on the RUL benchmark data set C-MAPSS. Summarizing, the models perform significantly worse for multiple operating conditions, while there is no notable difference between one and two fault modes. A clear trend towards deep learning, specifically recurrent architectures, was identified. We identified seven challenges, which can be generalized to RUL prognosis: (1) limited quantitative comparability between some of the models – which is not a challenge of the problem setting but rather of the reported research, (2) higher error rates for multiple operating conditions, and (3) a strong research focus on minimizing error rates, not so much addressing implementability aspects. Additionally, we identified challenges hindering the applicability of ML models, whether due to properties of the model, such as (4) interpretability and (5) quantification of model uncertainty, or due to properties of the data, such as (6) the need for domain adaptation and (7) the need for large amounts of labelled data. We argue that these challenges are in the way of successful applications of data-driven RUL prognosis. While the majority of the surveyed research addresses the minimization of error-rates, the identified challenges are yet to receive the same scientific attention.

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